GNN lecture

* Sample $x^{(l)} = [\ ]_{N\times 1}$

$$x^{(l+1)} = \sigma(W_l x^{(l)} + b^l)$$

$W_{0} = [\ \ ]_{3\times 2}$ $\quad x^{(0)} = [\ ]_{2\times 1}$ $\quad b^{(0)} = [\ ]_{3\times 1}$

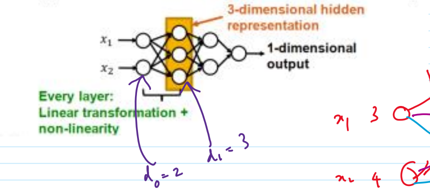
$W_0 \begin{bmatrix} 1 & 2 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}_{3\times 2} \begin{bmatrix} 3 \\ 4 \end{bmatrix}_{2\times 1} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}_{3\times 1}$

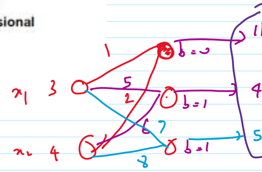
$\begin{bmatrix} 11 \\ 39 \\ 53 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 11 \\ 40 \\ 54 \end{bmatrix}$

Time l_n

$x^{(l+1)} = W_l x^{(l)} + b^{(l)}$

- Suppose x is 2-dimensional, with entries x_1 and x_2





$\sigma(11) \rightarrow 11$

$\sigma(40) \rightarrow 40 \quad \Rightarrow x^{(1)} = \begin{bmatrix} 11 \\ 40 \\ 54 \end{bmatrix}_{2\times 1}$

$\sigma(54) \rightarrow 54$

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 5 & 2 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$

$\mathbf{1}$

$b\ \mathbf{1}^T$

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* batch

$X = [\ |\ |\ |\ ]_{N\times 3}$

$= \begin{bmatrix} 3 & -1 & 0 \\ 4 & -2 & 1 \end{bmatrix}_{2\times 3}$

$$X^{(l+1)} = \sigma\left(W_l X^{(l)} + b^{(l)}\mathbf{1}^T\right)$$

Important correction!
During the lecture, ones row vector was used before the bias vector.
That was a mistake. It comes after the bias

$\Downarrow$

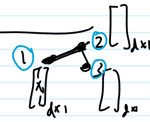
$\begin{bmatrix} 11 & \cdot & \cdot \\ 54 & \cdot & \cdot \\ 53 & \cdot & \cdot \end{bmatrix}_{3\times 3} + \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}_{3\times 3} \leftarrow b$

$\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$[1\ 1\ 1] = \mathbf{1}^T_{3\times 1}$

$\begin{bmatrix} | \\ b \\ | \end{bmatrix}_{N\times 1} [-\ 1\ -]_{1\times m} = \begin{bmatrix} | & | & \cdots & | \\ b & b & \cdots & b \\ | & | & & | \end{bmatrix}_{N\times m}$

GNN



$[\ ]_{d\times 1}$

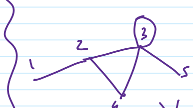
$d = 5$

$X = \begin{bmatrix} | & | & | \\ x_1 & x_2 & x_3 \\ | & | & | \end{bmatrix}_{5\times 3}$

$V = \{v_1, v_2, v_3\}$

$|V| = 3$

$X \in \mathbb{R}^{|V|\times d} \Rightarrow X \in \mathbb{R}^{5\times 3}$



$V = \{v_1, v_2, v_3, v_4, v_5\}$

$N(v) \Rightarrow N(v_3) = \{v_2, v_4, v_5\}$

* Node level representation

$$h_u^{(k)} = \sigma\left(W_{self}^{(k)} h_u^{(k-1)} + W_{neigh}^{(k)} \boxed{\sum_{v\in\mathcal{N}(u)} h_v^{(k-1)}} + b^{(k)}\right) \quad (7)$$

Simplified

where,

$$W_{self}^{(k)}, W_{neigh}^{(k)} \in \mathbb{R}^{d^{(k)}\times d^{(k-1)}}$$
$$b^{(k)} = b_{self}^{(k)} + b_{neigh}^{(k)}$$

In equation 7,
$W_{self}^{(k)}$: weight matrix of target node's (u) multi-layer perceptron (MLP),
$W_{neigh}^{(k)}$: weight matrix of MLP used with target node's neighborhood nodes ($\mathcal{N}(u)$),
$\sigma(.)$: activation function.

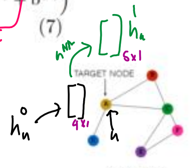


Fig. 1: GN

$v \in \{B, C, D\}$

$W_1 = [\ ]_{3\times 2}$ $\quad W_2 = [\ ]_{5\times 3}$ $\quad W_3 = [\ ]_{2\times 5}$

$x^0 = [\ ]_{2\times 1} \Rightarrow x^1 = W_1 x^0$

$x^2 = W_2 x^1 = W_2 W_1 x^0$

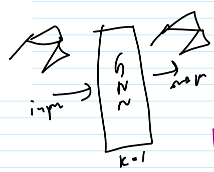
$x^3 = W_3 x^2 = \boxed{W_3 W_2 W_1} x^0$

$= [\ ]_{2\times 5} [\ ]_{5\times 3} [\ ]_{3\times 2} [\ ]_{2\times 1}$

$x^3 = [\ ]_{2\times 1}$

$W_3 W_2 W_1 = [\ ]_{2\times 2}$

$$h_u^k = \sigma\left(W_{self}^k h_u^{(k-1)} + \left(\sum_{v\in N(u)} W_{neigh}^k h_v^{(k-1)}\right) + b^k\right)$$



$W_{neigh} \in \mathbb{R}^{5\times 4}$

$$h_u^1 = \sigma\left(W_{self}^1 h_u^0 + \boxed{\left(\sum \boxed{W_{neigh}^1 h_v^0}\right)} + b^1\right)$$

Step 2

msg (Step 1)

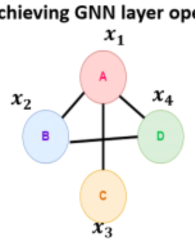
* Graph level Representation

Achieving GNN layer operation using Adjacency Matrix (a simple approach)

x_1

A B C D

Achieving GNN layer operation using Adjacency Matrix (a simple approach)



Graph G

	A	B	C	D
A	0	1	1	1
B	1	0	0	1
C	1	0	0	0
D	1	1	0	0

Adjacency matric for graph G

$$H^{(k)} = \sigma\left(AH^{(k-1)}W^{(k)}_{neigh} + H^{(k-1)}W^{(k)}_{self}\right)$$

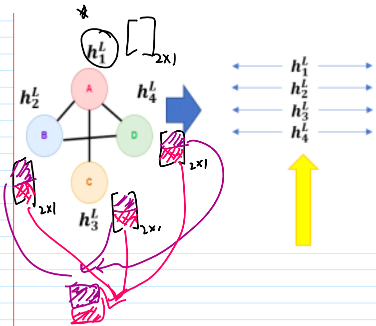
$$H^{(k)} = \sigma(\ \_\_\_ + \_\_\_ + \mathbf{1}\cdot b^{(k)})$$

Important correction
During the lecture, this was done in the wrong order

$$b^{(k)} = [\text{---} b \text{---}]_{1\times d^{(k)}}$$

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}_{d^{(k)}\times 1}$$

$$H^{(1)} = \sigma\left(AH^{(0)}W_n^{(1)} + H^{(0)}W_s^{(1)}\right)$$



$$H^{(0)} = \begin{bmatrix} \text{---} h_1 \text{---} \\ \text{---} h_2 \text{---} \\ \text{---} h_3 \text{---} \\ \text{---} h_4 \text{---} \end{bmatrix}_{4\times2}$$

node 1 →
node 2 →
node 3 →
4 →

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}_{4\times4} \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}_{4\times2} = \begin{bmatrix} 16 & 18 \\ 12 & \\ & \end{bmatrix}_{4\times2} \begin{bmatrix} \ \end{bmatrix}_{2\times D}$$

$$W^{(k)}_{neigh} = W^{(k)}_{self} = W^{(k)}_{shared}$$

* Weight Shared scenario

$$H^{(k)} = \sigma\left(AH^{(k-1)}W^{(k)}_{shared} + H^{(k-1)}W_{shared}\right)$$

$$* \quad H^{(k)} = \sigma\left((A+I)H^{(k-1)}W^{(k)}_{shared}\right)$$

$$I_{n\times n} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & \vdots \\ \vdots & & \ddots & \\ 0 & \cdots & & 1 \end{bmatrix}_{n\times n}$$